

Two worked domains, with different evidence timing

The analyses answer complementary questions; neither is a field maintenance trial.

A | Retrospective C-MAPSS

FD001-FD004

5 split levels × 10 streams

Retrospective; lock before aggregation



Completed analyses

Full crossing and endpoints

Preprocessing and metric rankings

Static and observed-prefix rules

B | Frozen battery study

NASA Li-ion data

16 of 34 units eligible

Frozen before training and test-label evaluation



Frozen comparison

Five training streams

Primary RMSE and LPR

Normalization and Ji denominator checks



Interpretation boundary

Finite-design contrasts and protocol sensitivity; no universal winner.

No optimized policy or field reliability claim.

No deployment-benefit claim.

N-CMAPSS DS02 is retained only as a Supplementary computational representation diagnostic.